

DEPARTMENT OF MATHEMATICS

BIRLA INSTITUTE OF TECHNOLOGY MESRA, RANCHI

MA107 Mathematics-II

Session: SP2021

Tutorial - 2 (Module II)

1. Determine the singular points of the following differential equations and examine whether they are regular or not

a) $(x^2 + 4)(x - 3)^3 y'' + (x - 3)^3 y' + y = 0$

b) $y'' + \frac{1}{x} y' + \left(1 - \frac{1}{x^2}\right) y = 0$

c) $x^2 y'' + y' + 3xy = 0$

d) $x^2 y'' + xy' + (x^2 - p^2)y = 0$

e) $(1 - x^2)y'' - 2xy' + n(n + 1)y = 0$

f) $x^3(x - 1)y'' - 2(x - 1)y' + 3xy = 0$

g) $x^2(1 - x^2)y'' + \frac{2}{x}y' + 4y = 0$

2. Consider the equation $y'' + xy' + y = 0$.

- a) Find its general solution $y = \sum a_n x^n$ in the form $y = a_0 y_1(x) + a_1 y_2(x)$, where $y_1(x)$ and $y_2(x)$ are power series.

- b) Show that $y_1(x)$ is the series expansion of $e^{-x^2/2}$, use this fact to find the second independent solution.

3. Find the solution of the following equations in power series

a) $xy' = 3(y + 1)$

b) $y'' - 3y' + 2y = 0$

c) $2x(1 - x)y'' + (1 - x)y' + 3y = 0$

4. Use Frobenius method to find the solutions of the following equations

a) $xy'' + y' + xy = 0$

b) $xy'' + 3y' + 4x^3y = 0$

c) $x(2 + x^2)y'' - y' - 6xy = 0$

d) $4xy'' + 2y' + y = 0$

e) $x(1 - x)y'' + 4y' + 2y = 0$

f) $9x(1 - x)y'' - 12y' + 4y = 0$

5. Express the following expression in terms of Legendre polynomials

a) $2 - x + 3x^2$

b) $x^3 - 7x^2 - 28x - 15$

6. Prove that

a) $P_n(1) = 1$

b) $P_n(-1) = (-1)^n$

c) $P'_n(1) = \frac{1}{2}n(n + 1)$

d) $P'_n(-1) = \frac{1}{2}(-1)^n[n(n + 1)]$

e) $P_n(-x) = (-1)^n P_n(x)$

7. Prove that $\int_{-1}^1 P_n(x)(1 - 2xt + t^2)^{-1/2} dx = \frac{2t^n}{2n + 1}$.

8. (**Recurrence relations for $P_n(x)$**) To show that

1. $(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x), n \geq 1$
2. $nP_n(x) = xP'_n(x) - P'_{n-1}(x)$
3. $(2n+1)P_n(x) = P'_{n+1}(x) - P'_{n-1}(x)$
4. $(n+1)P_n(x) = P'_{n+1}(x) - xP'_n(x)$
5. $(1-x^2)P'_n(x) = n(P_{n-1}(x) - xP_n(x))$
6. $(1-x^2)P'_n(x) = (n+1)(xP_n(x) - P_{n+1}(x))$
7. $(2n+1)(x^2-1)P'_n(x) = n(n+1)(P_{n+1}(x) - P_{n-1}(x))$ (**Beltrami's result**)

9. Show that when n is an integer $J_{-n}(x) = (-1)^n J_n(x)$

10. Prove that

$$\text{a) } J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \cos(x) \quad \text{b) } J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin(x) \quad \text{c) } [J_{1/2}(x)]^2 + [J_{-1/2}(x)]^2 = \frac{2}{\pi x}$$

11. Write the general solution of the following equations (in terms of Bessel functions)

$$\text{a) } x^2 y'' + xy' + (x^2 - 25)y = 0 \quad \text{b) } x^2 y'' + xy' + (x^2 - \frac{9}{16})y = 0$$

12. (**Recurrence relations for $J_n(x)$**) To show that

$$\begin{array}{ll} \text{a) } \frac{d}{dx} \{x^n J_n(x)\} = x^n J_{n-1}(x) & \text{b) } \frac{d}{dx} \{x^{-n} J_n(x)\} = -x^{-n} J_{n+1}(x) \\ \text{c) } J'_n(x) = J_{n-1}(x) - (n/x)J_n(x) & \text{d) } J'_n(x) = (n/x)J_n(x) - J_{n+1}(x) \\ \text{e) } J_{n-1}(x) + J_{n+1}(x) = (2n/x)J_n(x) & \text{f) } 2J'_n(x) = J_{n-1}(x) - J_{n+1}(x) \end{array}$$

13. Evaluate $\int x^3 J_3(x) dx$